Advanced Programming Techniques

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C++ Libraries
Boost Socket

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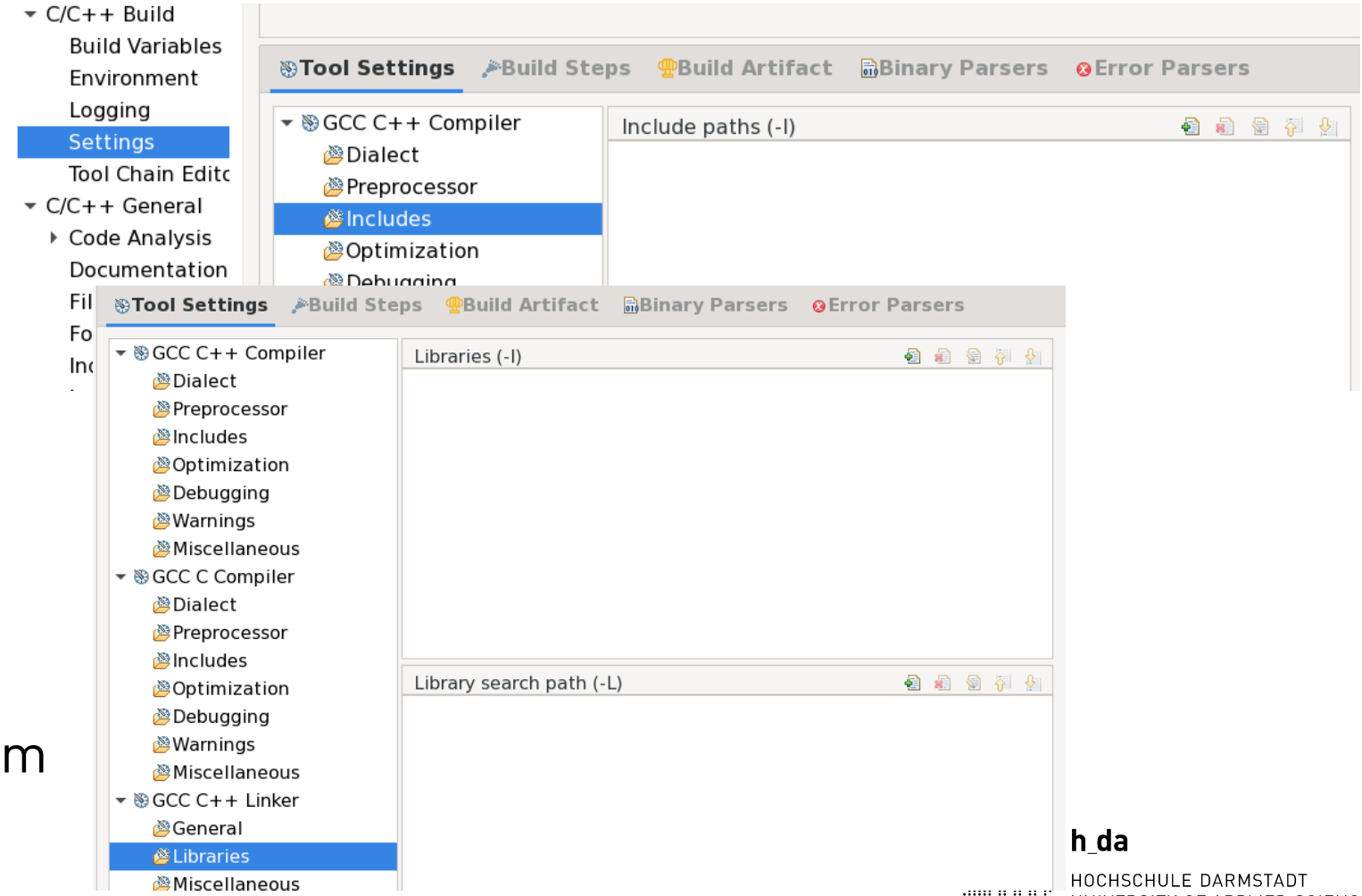
More Libraries

- **The C++ Standard library does not cover everything**
- **Notably no classes for network connections**
- **Additional libraries are available**
  - e.g. https://github.com/fffaraz/awesome-cpp
- **Examples**
  - Boost: well known, large, covers a lot of topics
  - Poco: focused on developing network applications
  - FLTK: GUI library

Adding a library

- **Structure**
  - A library consists of
    - header files
    - precompiled code (a.k.a “library”)
- **Adding to project**
  - Copy all files into project
  - Configure in Eclipse
- **Use “packaged” library**
  - Install in Operating system
  - Configure in Eclipse



Networking

- **Dominated today by the Internet-Protocol (IPv4, IPv6)**
- **Basic concepts**
  - Network participants (“Computers”) have IP-addresses
    - Example IPv4: 172.16.254.1 (32 bits denoted as decimal values of four 8-bit groups)
    - Example IPv6: 2001:0db8:85a3:0000:0000:8a2e:0370:7334 (128 bits denoted as hex values of 8 groups with 16 bits each)
    - Luckily a service (Domain Name Service) maps human readable names (e.g. www.h-da.de) to IP-addresses
  - Data is exchanged between communication endpoints
    - An endpoint consists of an IP-address and a port number
  - Servers “listen” for connections, clients connect to servers
  - A connection is a pair of two end points

Simple service example

- **Daytime Service (RFC 867)**
  - Listens on port 13
  - Upon connection, returns string with time information
- **Public servers**
  - `time-a-g.nist.gov`
  - `time-b-g.nist.gov`
  - `time-c-g.nist.gov`
  - `time-c-g.nist.gov`

Boost socket

- Part of the Boost Asio (asynchronous i/o library)
  - https://www.boost.org/doc/libs/1_70_0/doc/html/boost_asio/reference.html
- Communication endpoint modeled as “socket iostream”
- Boost likes namespaces
  - Full class name: `boost::asio::ip::tcp::iostream`
- Socket can be created and then connected or connected upon creation
- After successful connection, socket can be used like an **`std::iostream`**

Live Coding

Aside: namespaces

- **C++ namespaces**
  - Often used, never explained
  - A namespace is a scope in which names have to be unique
    - Example:
      - In Germany street names have to be unique within a town
      - Streets with the same name may exist in different towns
      - → A town is a namespace for street names
  - The standard library uses namespace `std` for everything
  - We have used the global namespace for our classes

Aside: namespaces

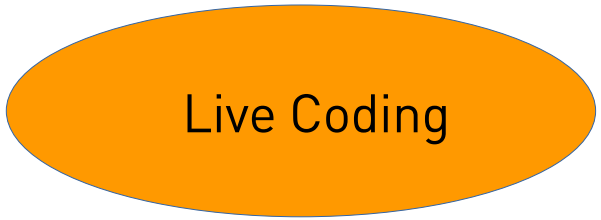
- “Putting” (defining) something in a namespace

```
namespace myNameSpace {

class Test {
};

}
```

- Referencing something from a namespace
  - With prefix: `myNameSpace::Test`
  - With using: `using namespace myNameSpace`
- Rules
  - Never use `using` in header files (you cannot “undo” it)
  - Avoid using `using` in Implementation files (prefix indicates origin)

Time for machine processing

- **Time Protocol (RFC 868)**
  - Have a look: https://tools.ietf.org/html/rfc868
- **Steps**
  - Create connection
  - Convert 4 bytes to 32 integer
  - Convert to “time”



Working with time/date is difficult

- **How can we model date/time**
  - Class with data members day, month, year, hour minute?
    - Many items
    - Uses unnecessary amount of storage space
    - Calculating duration is a nightmare
      - How many seconds from 24.01.1990 13:42 to 04.12.2019 17:45?
  - The Unix way: Seconds since 1.1.1970 0:00 UTC ("the epoch")
    - Hard to understand for humans
    - Conversion to "readable" time isn't easy
      - Seconds: `secondsSince % 60`, minutes: `secondsSince / 60 % 60`, hours: `secondsSince / 3600 % 24`, day of month: … now it gets difficult
    - → we need another library
    - Be aware of range, or "Reboot your dreamliner" (https://arstechnica.com/information-technology/2015/05/boeing-787-dreamliners-contain-a-potentially-catastrophic-software-bug/)

C++11 chrono (originally from Boost)

- **Clocks**
  - Used to obtain absolute or relative time
    - `system_clock`
    - `steady_clock`
    - `high_resolution_clock`
- **Time point**
  - Duration of time that has passed since epoch
- **Duration**
  - Span of time with unit
- **Plus support from C library (`std::time_t`)**

`std::time` and related functions

- **`std::time_t` is used to represent a point in time**
  - Implemented almost always as an integral type that holds the number of seconds since 00:00, Jan 1 1970 UTC
  - Can be converted to `std::tm` which represents the time "as we know it" (day, month, year, hour, minute, second), either by `localtime` or `gmtime`
    - `std::tm` can be converted to `std::time_t` using `std::mktime`
  - Can be converted to human readable form using `std::put_time`
    - See code examples in cppreference.com